

PALLETI VIRAJA

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EDUCATION

Shiv Nadar University, Delhi, NCR

Aug 2024 – Present

B.Tech in Computer Science and Engineering, Second Year, GPA: 7.8

Narayana Junior College, Hyderabad, 12th Grade: 95.2%

May 2024

JEE Mains: 96.5 Percentile

EXPERIENCE

Research Intern - IIIT Hyderabad - Language Technologies Research Centre (LTRC)

May - Aug 2026

Advisor Prof. Chiranjeevi Yarra

Developing an automatic prosodic phrase classifier for English intonational categories, with a custom-annotated ToBI dataset and full speech processing pipeline

PROJECTS AND PAPERS

Hermes: Low-Communication Distributed LLM Training Paper *(In works of being published)*

A systems paper implementing GPU-fused gradient compression via custom CUDA kernels, reducing compression **latency from 160ms to 3.5ms (46x)** and total bandwidth by up to 13,000x over standard DDP

Introduces heterogeneity-aware sharding enabling LLM training across mixed hardware over commodity WiFi

BlindSpot: Hardware Navigation System for Visually Impaired

Secured 1st place among 40+ teams; deployed YOLOv8 on Raspberry Pi using NCNN for ARM-optimised real-time inference, with directional haptic feedback via a companion mobile app

- *Tech Stack*: Python, YOLOv8, NCNN, Raspberry Pi, Flask, Svelte, OpenCV

Refract: Style Adaptive PDF Assistant *(In Progress)*

An intelligent document assistant that ingests PDFs and dynamically adapts explanations to the user's own writing style using transformer-based semantic similarity

- *Tech Stack*: Python, PyPDF2, HuggingFace Transformers, Sentence Transformers, pyttsx3

Dungeon Master: Interactive RPG

Stateful RPG engine managing persistent game state across turns using LangGraph, with tool-calling for game mechanics and human-in-the-loop interrupts for turn-based gameplay

- *Tech Stack*: Python, LangGraph, LLM Prompt Engineering

Hyperlocal AQI Forecaster: Delhi PM2.5 Prediction Dashboard

Deployed Plotly Dash dashboard forecasting 24h-ahead PM2.5 for 3 Delhi monitoring stations; XGBoost beats persistence baseline by up to 12% MAE at volatile stations after diagnosing and correcting a 20-unit train/test overfit gap via targeted regularisation

- *Tech Stack*: Python, pandas, XGBoost, scikit-learn, Plotly Dash, Render

LEADERSHIP & ACTIVITIES

- **Winner of Hackathon 2026** – Hackdata
- *Secretary of the Treasury* – Shiv Nadar University Student Council 25-26
- Top 10 teams from Shiv Nadar – Smart India Hackathon 2025
- ACM SNIoE: Data Analytics team member
- Astronautics and Aeronautics Society, SNIoE: Technical team
- Google Developers Group, SNIoE: AI/ML team member

SKILLS & INTERESTS

Languages: C++, Python, C, Java, JavaScript

Systems & Infrastructure: SQL & NOSQL database, AWS, CUDA, Git

AI/ML: PyTorch, Scikit-learn, HuggingFace, LangChain, LangGraph

Web/Backend: Flask, React, Node

Tools: Git, Linux, Raspberry Pi

Relevant Coursework: Data Structures, Computer Architecture, Object-Oriented Programming, Introduction to LLMs, Introduction to Machine Learning